

APPROXIMATE DISTRIBUTIONS FOR LARGEST AND FOR SMALLEST OF A SET OF INDEPENDENT OBSERVATIONS

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ABSTRACT

Consider n independent univariate observations with possible different distributions. Let X_n and X_1 denote the largest and smallest observations, respectively, while $\bar{F}(x;n)$ is the arithmetic average of the cumulative distribution functions for the individual observations. Approximate expressions, also sharp upper and lower bounds, are developed for $P(X_n \leq x)$ and $P(X_1 \leq x)$ in terms of n , $\bar{F}(x;n)$, and the level of generality. Various levels of generality are considered, including a form of complete generality. The approximate expressions are near the bounds for all levels of generality if $P(X_n \leq x) \geq .8$, or if $P(X_1 \leq x) \leq .2$, respectively. All possible distributions and all $n \geq 1$ are allowable. Suitable estimation of $\bar{F}(x;n)$ provides estimates of $P(X_n \leq x)$ and $P(X_1 \leq x)$. In particular, for continuity in the pertinent tail of $\bar{F}(x;n)$, a type of asymptotic approximation that involves at most three parameters is sometimes usable. Then, the problem is simplified to estimating the parameters. Approximate confidence regions and tests are obtained for sufficiently extreme upper and lower percentage points of $\bar{F}(x;n)$. Also, for continuity in the pertinent tail, approximate one-sided tolerance intervals of the form (X_n, ∞) , $(-\infty, X_1)$ are developed for the tails of $\bar{F}(x;n)$. Finally, tests of $\bar{F}(x;n) \equiv F_0(x)$, completely specified, are developed for x in the pertinent tail.

1. Introduction and discussion

There is often interest in whether the largest observation of a set of n independent observations is unusually large, or whether the smallest is unusually small. Rather accurate approximate probability expressions of a convenient type can be developed for relations of this kind when the probability values are small (say, at most .2). These expressions are applicable for all $n \geq 1$ and for greatly different distributions (arbitrarily different, in one case) for the individual observations.

For a given x value, r of the cumulative distribution functions (cdf's) are required to be at most equal to $\bar{F}(x;n)$ for the case of $P(X_n \leq x)$, and r cdf's are at least equal to $\bar{F}(x;n)$ for $P(X_1 \leq x)$. The value of r identifies the level of generality (complete generality for $r = 1$).

More specifically, let X_n and X_1 be the largest and smallest observations, respectively. Approximate expressions are developed for $P(X_n \leq x)$ and $P(X_1 \leq x)$

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that are rather accurate if $P(X_n > x) \leq .2$, or if $P(X_1 \leq x) \leq .2$. They are very accurate if .2 is replaced by .15. The level of accuracy increases as r increases. These approximate expressions are arithmetic averages of the sharp upper and lower bounds and are functions of n , r , and $\bar{F}(x;n)$, where $\bar{F}(x;n)$ is the arithmetic average of the cdf's for the individual observations. The expression and bounds for $P(X_n \leq x)$ are meaningful only for x such that $(n+1-r)(1-\bar{F}(x;n)) < 1$, and those for $P(X_1 \leq x)$ only for x such that $(n+1-r)\bar{F}(x;n) < 1$. Their principal usefulness is for situations where $P(X_n > x) \leq .4$ and $P(X_1 \leq x) \leq .4$; these relations are always satisfied for x such that $(n+1-r)(1-\bar{F}(x;n)) \leq .4$ and $(n+1-r)\bar{F}(x;n) \leq .4$, respectively.

A desirable feature of the approximate expressions developed is the ease with which the $\bar{F}(x;n)$ for a new set of n observations can be estimated from previous observations with this same arithmetic average for their cdf's. Given n , r , and a suitable estimate of $\bar{F}(x;n)$, the values of $P(X_n \leq x)$ and $P(X_1 \leq x)$ that would be obtained for the n new observations are estimated for x such that, say, $(n+1-r)(1-\bar{F}(x;n)) \leq .2$ and $(n+1-r)\bar{F}(x;n) \leq .2$, which imply that $P(X_n > x) \leq .2$ and $P(X_1 \leq x) \leq .2$, respectively.

An obvious way to estimate $\bar{F}(x;n)$ is by the empirical cdf of the set of past observations. Unless the number of past observations is very much greater than n , however, a noticeably discontinuous estimate (of a step function nature) is obtained.

Suitable estimates of a continuous nature, without the amount of **past data** being exceedingly large compared to n , can often be obtained when $\bar{F}(x;n)$ is continuous (or very nearly so) in the pertinent tail and n is quite large. First, consider the case of X_n and use of an analogy with results for the case of a random sample (for example, see Walsh, 1964 and 1965). Approximation in the tail of $\log_e (1-\bar{F}(x;n))$ by an expression of the form $-ax+b$, ($a > 0$), or one of the other two forms occurring as an "asymptote" for X_n in the sample case (see Gumbel, 1958), should often be possible. Use of one of these three forms of approximation reduces the problem of estimating $P(X_n \leq x)$ to estimation of two or three parameters. A weighted type of least-squares procedure can be used for estimation of these parameters.

Similar considerations apply to X_1 when n is large and $\bar{F}(x;n)$ is continuous in its lower tail. The three forms of approximation to $\log_e \bar{F}(x;n)$ are those occurring in an "asymptote" of X_1 for the sample case. The parameters in the form of approximation used can be estimated by a least-squares procedure similar to that developed for X_n .

The cdf $\bar{F}(x;n)$ plays an important role for investigations involving order statistics of sets of independent observations (also see Walsh, 1959). Thus, investigation of its properties can be of interest. Several kinds of procedures are given for investigating $\bar{F}(x;n)$ in a specified tail.

Extreme upper percentage points of $\bar{F}(x;n)$, also extreme lower percentage points, can be investigated by use of X_n or X_1 , respectively. One-sided confidence intervals and tests are easily developed when $\bar{F}(x;n)$ is continuous (or very nearly so) at the percentile considered. These intervals and tests have rather accurate

probability levels if the upper percentage points correspond to values of x such that $(n+1-r)(1-\bar{F}(x;n)) \leq .2$, and the lower percentage points to x such that $(n+1-r)\bar{F}(x;n) \leq .2$. Even for percentiles this extreme, the probability levels are only bounded, instead of rather accurately determined, when $\bar{F}(x;n)$ is discontinuous at the percentile considered.

One-sided tolerance intervals for $\bar{F}(x;n)$ of the less standard but useful forms (X_n, ∞) and $(-\infty, X_1)$ can be obtained. The interest is whether more than a specified (small) amount of the probability of $\bar{F}(x;n)$ is covered in the tail considered. Continuity of $\bar{F}(x;n)$ in the pertinent tail is assumed and the lower bound values are used for the probabilities of the tolerance interval relations.

Finally, tests are easily developed for the two null hypotheses: $\bar{F}(x;n) \equiv \bar{F}_0(x)$, completely specified, in the upper tail; $F(x;n) \equiv \bar{F}_0(x)$ in the lower tail. Tests for the upper tail use X_n , and tests for the lower tail are based on X_1 .

It is to be noted that a distribution exists such that X_n is the largest order statistic for a sample of size n from this distribution. Also, a distribution occurs for which X_1 is the smallest value in a sample size n . The cdf corresponding to X_n is the geometric mean of the cdf's for the separate observations. The cdf corresponding to X_1 is unity minus the geometric mean of the probabilities of exceeding x for the individual observations. However, their multiplicative definitions result in obscure interpretation of the properties for these distributions, except when they can be approximately expressed in terms of $\bar{F}(x;n)$. Also, distributions that are not easily related arise for different order statistics. An additional advantage of expressing the results in terms of $\bar{F}(x;n)$ is that the distributions for X_n , X_1 and, in fact, all of the order statistics can be expressed (approximately) in terms of n and $\bar{F}(x;n)$ for most situations. Moreover values in excess of .8 for $P(X_n \leq x)$ and less than .2 for $P(X_1 \leq x)$ are determined with good accuracy when $F(x;n)$ is used. Finally, experimental investigation (estimation, etc.) of the arithmetic average of the cdf's is much easier than investigation of their product.

The following section contains the bounds and approximate expressions for $P(X_n \leq x)$ and $P(X_1 \leq x)$, along with some derivations. Least-squares estimation of parameters for the asymptotic distributions is considered in the next section. Then, there is a section devoted to confidence intervals and tests for upper or lower percentage points of $\bar{F}(x;n)$. The next to last section is concerned with tolerance intervals for $\bar{F}(x;n)$, and the last section contains tests of whether $\bar{F}(x;n)$ equals a completely specified cdf in a stated tail.

2. Bounds and approximate expressions

When $(n+1-r)[1-\bar{F}(x;n)] < 1$, sharp upper and lower bounds for $P(X_n < x)$ are given by

$$\{1 - (n+1-r)[1-\bar{F}(x;n)]\} F(x;n)^{r-1} \leq P(X_n \leq x) \leq F(x;n)^n.$$

Sharp upper and lower bounds for $P(X_1 \leq x)$ are provided by

$$1 - [1-\bar{F}(x;n)]^n \leq P(X_1 \leq x) \leq 1 - [1-(n+1-r)\bar{F}(x;n)][1-\bar{F}(x;n)]^{r-1}$$

when $(n+1-r)\bar{F}(x;n) < 1$.

The approximate expression for $P(X_n \leq x)$ is denoted by $A_n(\bar{F}(x;n);n,x)$ and equals the arithmetic average of the sharp upper and lower bounds for $P(X_n \leq x)$. The value of $A_n(\bar{F}(x;n);n,r)$ is near both bounds when $(n+1-r)(1-\bar{F}(x;n)) \leq .2$.

The approximate expression for $P(X_1 \leq x)$ is denoted by $A_1(\bar{F}(x;n);n,r)$ and is the arithmetic average of the sharp upper and lower bounds. The value of $A_1(\bar{F}(x;n);n,r)$ is near both bounds when $(n+1-r)\bar{F}(x;n) \leq .2$.

The sharp upper and lower bounds are derived only for $P(X_n \leq x)$. A similar derivation yields sharp upper and lower bounds for $P(X_1 > x) = 1 - P(X_1 \leq x)$, and thus for $P(X_1 \leq x)$.

The sharp lower bound is derived first. Let $F_i(x)$ denote the cdf for the i -th of the n independent observations. Then, $P(X_n \leq x)$ can be expressed as

$$\begin{aligned} \prod_{i=1}^n F_i(x) &= \exp\left\{\sum_{i=1}^n \log_e \{1 - [1 - F_i(x)]\}\right\} \\ &= \exp\left\{\sum_{i=1}^n \sum_{j=1}^{\infty} [1 - F_i(x)]^j / j\right\} \\ &= \exp\left\{-\sum_{j=1}^{\infty} j^{-1} \sum_{k=0}^j \binom{j}{k} [1 - \bar{F}(x;n)]^{j-k} \sum_{i=1}^n [\bar{F}(x;n) - F_i(x)]^k\right\}. \end{aligned}$$

For $(n+1-r)[1 - \bar{F}(x;n)] < 1$ and $k \leq 2$,

$$(1) \quad \sum_{i=1}^n [\bar{F}(x;n) - F_i(x)]^k$$

is largest when all but r of the $F_i(x)$ are unity, $r-1$ of them equal $\bar{F}(x;n)$, and the remaining one is such that their arithmetic average is $\bar{F}(x;n)$. This is readily verified by considering the cases of $k = 2, 3$ and the relative relationships for larger k . These choices for the $F_i(x)$ imply that the remaining $F_i(x)$ equals $1 - (n+1-r)(1 - \bar{F}(x;n))$. For $k = 0$ and 1 , the values of (1) are the same for all possible choices of the $F_i(x)$. Thus, the maximizing of (1) for $k \geq 2$ minimizes the argument of the exponential. This implies that the lower bound for $P(X_n \leq x)$ is

$$\{1 - (n+1-r)[1 - \bar{F}(x;n)]\} \bar{F}(x;n)^{r-1},$$

the value for the choice of the $F_i(x)$ that maximizes (1) for $k \geq 2$.

The sharp upper bound for $P(X_n \leq x)$ is obtained by noticing that, since the geometric mean at most equals the arithmetic mean, the probability is maximized when all the $F_i(x)$ equal $\bar{F}(x;n)$. Thus, this upper bound is $\bar{F}(x;n)^n$. This upper bound and its basis are applicable for all r .

For $r = 1$, alternative proofs for both bounds could be developed from the results of Hoeffding, (1956).

3. Estimation of Asymptotic parameters

First, some further remarks are made about the justification for applying approximations of the three types that occur for the random sample case to $\log_e(1 - \bar{F}(x;n))$ and to $\log_e \bar{F}(x;n)$. For n large but fixed, $\bar{F}(x;n)$ has some form. If this form is of the type that, for n this large, would permit the approximation for the sample case, the approximation also applies to this case. Fortunately when n is very large a rather broad class of cdf's are of a suitable type. Similar

to the random sample case, there are difficulties in deciding whether an approximation is usable for a practical application with given n but unknown $\bar{F}(x;n)$ and, if so, which of the three types to use. To solve this difficulty even for the sample case is beyond the scope of this paper. It is assumed that n and $\bar{F}(x;n)$ are such that an identified approximation is usable.

Estimates are developed only for the case of X_n , but a similar procedure yields estimates for the case of X_1 . The estimation procedure given is of a weighted least-squares nature and has only a curve-fitting basis.

The other two forms of parametric asymptotic distributions can be transformed into that where $\log_e [1 - \bar{F}(x;n)]$ is approximated by $-ax + b$ for the case of X_n (and where the approximation is of the form $ax + b$ for the case of X_1 , with $a > 0$). These transformations are accomplished by suitable changes of variable (for example, see Walsh, 1965). The change of variable involves an additional parameter for the third asymptote (and perhaps for the second asymptote) but a modification allows a , b , and this parameter to all be estimated by the weighted least-squares procedure.

Values of x in the neighbourhood of $\bar{F}(x;n) = 1 - k/(n+1-r)$ are considered to be of greatest interest (for $k \leq .4$, say, and suitably chosen). As $\bar{F}(x;n)$ deviates from this value, the interest decreases (perhaps differently in each direction). The relative interest is very small (perhaps zero) when $\bar{F}(x;n)$ is less than $1 - 1/(n+1-r)$. This relative interest can be quantitatively expressed in terms of a weighting function $W(\bar{F}(x;n))$.

Suppose that the previous data consist of several independent sets of independent observations (whose cdf's average, for each set, approximately to $\bar{F}(x;n)$ in the upper tail). Pool these observations into a single set of size m (where m should be much larger than n for good estimation). Let $G_m(x)$ denote the empirical cdf of these observations. First consider the case where only the two parameters $\underline{a}$ and $\underline{b}$ are to be estimated. The estimates are the values which minimize the sum

$$(2) \quad \sum W[G_m(x)] \{ \log_e [1 - G_m(x)] + ax - b \}^2.$$

x = observed values.

They are easily determined by setting the partial derivatives with respect to $\underline{a}$ and $\underline{b}$ equal to zero and simultaneously solving these two equations for $\underline{a}$ and $\underline{b}$.

Additional effort is required when the third asymptote was transformed. Then, the estimates for $\underline{a}$ and $\underline{b}$ provide estimates for two parameters of the third asymptote but a third parameter is also involved (occurs explicitly in the change of variable). This parameter is estimated iteratively, with its value being that which minimizes the value of (2) that is first minimized with respect to $\underline{a}$ and $\underline{b}$. That is, starting with an initial guess, values are considered for this third parameter and the corresponding minimum values of (2) with respect to $\underline{a}$ and $\underline{b}$ are determined. Ultimately, perhaps using interpolation among the cases already considered, a close approximation to the minimizing value of the third parameter is determined. This approach is also usable when the

second asymptote is considered to have three parameters rather than the usual two. Of course, the estimates for $\underline{a}$ and $\underline{b}$ are those corresponding to the minimizing value of the third parameter.

4. Investigation of extreme percentiles

Let θ_p denote the 100p percent point of $\bar{F}(x;n)$. The principal interest is in the case where $p \geq 1 - .2/(n+1-r)$ for the upper tail and the case where $p \leq .2/(n+1-r)$ for the lower tail.

First, consider cases where $\bar{F}(x;n)$ is continuous (or very nearly so) at θ_p . For $p \geq 1 - .2/(n+1-r)$,

$$P(X_n \leq \theta_p) \doteq A_n[p;n,r].$$

Thus, the interval (X_n, ∞) is a one-sided confidence interval for θ_p with confidence coefficient approximately equal to $A_n(p;n,r)$ and bounds near this value. For $p \leq .2/(n+1-r)$,

$$P(X_1 > \theta_p) \doteq 1 - A_1[p;n,r]$$

provides $(-\infty, X_1)$ as an approximate confidence interval for θ_p .

Other ranges of p such that $p > 1 - 1/(n+1-r)$, or $p < 1/(n+1-r)$, could be considered in obtaining one-sided confidence intervals. Even for continuity at θ_p , however, the bounds on a confidence coefficient are only moderately close together for $1-p \leq .5/(n+1-r)$ using X_n , or for $p \leq .5/(n+1-r)$ using X_1 . They are rather far apart in other cases.

If $\bar{F}(x;n)$ is not continuous at θ_p , the values of $P(X_n \leq \theta_p)$ and $P(X_1 \leq \theta_p)$ are at least equal to the values for continuity at θ_p . The values of $P(X_n < \theta_p)$ and $P(X_1 < \theta_p)$ are at most equal to the values of $P(X_n \leq \theta_p)$ and $P(X_1 \leq \theta_p)$, respectively, for continuity at θ_p .

One-sided tests of the form $\theta_p = \theta_0$, with θ_0 a stated number, can be obtained in the usual straightforward manner. Also, X_n is an approximate median estimate of θ_p for p such that $A_n(p;n,r) = 1/2$, and X_1 is an approximate median estimate of θ_p for p such that $A_1(p;n,r) = 1/2$. Here, "approximate" implies that the probability of the estimate exceeding, or being below, θ_p does not differ much from $1/2$.

5. One-sided tolerance intervals

One-sided tolerance intervals are developed for the upper and the lower tail of $\bar{F}(x;n)$. They have the forms (X_n, ∞) and $(-\infty, X_1)$. Only cases where $\bar{F}(x;n)$ is continuous (or very nearly so) in the pertinent tail are considered.

Given any distribution function $\bar{F}(x;n)$ the probability included in the random interval (X_n, ∞) equals $1 - \bar{F}(X_n;n)$, and the probability in $(-\infty, X_1)$ equals $\bar{F}(X_1;n)$. Thus, the probability that (X_n, ∞) covers at least 100p percent probability in the upper tail of $\bar{F}(x;n)$ is

$$P[1 - \bar{F}(X_n;n) > p] = P(X_n \leq \theta_{1-p})$$

and hence is at least equal to $(1 - (n+1-r)p)(1-p)^{r-1}$ and at most equal to $(1-p)^n$. The eligible values of p are such that $p < 1/(n+1-r)$, with principal interest in cases where $p < .2/(n+1-r)$. As p decreases, the probability that (X_n, ∞) contains at

least 100p percent probability in the upper tail of $\bar{F}(x;n)$ increases, and approaches unity as the limit.

Similarly, the probability that $(-\infty, X_1)$ covers at least 100p percent probability in lower tail of $F(x;n)$ equals

$$P[\bar{F}(X_1;n) > p] = 1 - P(X_1 \leq p).$$

This probability is at least equal to $[1-(n+1-r)p](1-p)^{r-1}$ and at most equal to $(1-p)^n$, where $p < 1/(n+1-r)$ and principal interest is in $p < .2/(n+1-r)$.

6. Hypothesis of specified distribution

Let $\bar{F}(x;n)$ be completely specified as $F_0(x)$ in the upper tail, or the lower tail, by the null hypothesis. The upper tail is for x such that $F_0(x) > 1-1/(n+1-r)$ the lower tail is for x such that $F_0(x) < 1/(n+1-r)$.

Suppose that the principal interest is in disagreement between $\bar{F}(x;n)$ and $F_0(x)$ in the upper tail; that is, for x such that $F_0(x) > p$, where p is an attainable value for $F_0(x)$ and, under the null hypothesis,

$$P[F_0(X_n) > p] = P[X_n > \theta_p]$$

has a small value that is suitable for a significance level. Then, the test that rejects $\bar{F}(x;n) = F_0(x)$ in the upper tail if and only if $F_0(X_n) > p$ has a suitable significance level and emphasizes alternative hypotheses where $\bar{F}(x;n)$ is less than $F_0(x)$ in the upper tail.

From knowledge of $F_0(x)$ the bounds derived could be used to obtain approximately the significance level $P[F_0(X_n) > p | \bar{F}(x;n) = F_0(x)]$. For small values (say, at most .2), however, p is so near unity that the significance level is approximately $1-A_n[p;n,r]$, with bounds that are close together.

A similar test is easily developed for the lower tail and X_1 used. Then, $\bar{F}(x;n) = F_0(x)$ in the lower tail is rejected if and only if $F_0(X_1) \leq p$. This test emphasizes alternatives where $\bar{F}(x;n)$ exceeds $F_0(x)$ in the lower tail and has a significance level approximately equal to $A_1[p;n,r]$.

REFERENCES

- GUMBEL, E.J. (1958). *Statistics of Extremes*, Columbia University Press.
- HOEFFDING, W. (1956). On the distribution of the number of successes in independent trials, *Ann. Math. Statist.*, **27**, 713-721.
- WALSH, J.E. (1959). Definition and use of generalized percentage points, *Sankhyā*, **21**, 281-288.
- WALSH, J.E. (1964). Approximate distribution of extremes for nonsample cases, *Jour. Amer. Stat. Assoc.*, **59**, 429-436.
- WALSH, J.E. (1965). *Handbook of Nonparametric Statistics, II*, D. Van Nostrand Co.